

Physics

CSIT — 1st Semester — 2078 — Answer Sheet
Subject Code: PHY118

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Discuss electrical conductivity of semiconductors. Derive expression for conductivity in terms of impurity ionization energy. Give a plot to discuss 'Theoretical temperature dependence of the electrical conductivity of an impurity semiconductor'.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Describe processes involved in the fabrication of integrated circuits include epitaxial growth, oxidation, oxide removal and pattern definition, doping (impurities in the Si), and interconnection of components.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Find expression for force on a current-carrying wire in a magnetic field to find the force experienced by a single charge.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Give spectrum of Hydrogen atom and discuss its lines.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Give a brief account of Krongi-Penny model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe the term 'space quantization'.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. A given spring stretches 104 m when a force of 20N pulls on it. A 2-kg block attached to it on a frictionless surface is pulled to the right 0.2 m and released.(a) What is frequency of oscillation of the block?(b) What are the velocity and acceleration when $x = 0.12\text{m}$, on the block's first passing this point?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. A proton is moving with a velocity $\mathbf{v} = (3 \times 10^5\mathbf{i} + 7 \times 10^5\mathbf{k})\text{m/sec}$ in a region where there is a magnetic field $\mathbf{B} = 0.4\mathbf{j}\text{ T}$. Find the force experienced by the proton.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. In neutron spectroscopy a beam of mono-energetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If the separation between the atomic planes of the beryllium crystal is $0.732\text{ \AA}$, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength $0.1\text{ \AA}$?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. (a) How many atomic states are there in hydrogen with $n=3$?(b) How are they distributed among the sub shells? Label each state with the appropriate set of quantum numbers n, l, m_l, m_s .(c) Show that the number of states in a shell, that is, states having the same n , is given by $2n^2$. (Hint: $1+2+3+\dots+n=n(n+1)/2$).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The density of aluminum is 2.70 g/cm^3 and its molecular weight is 26.98 g/mole .(a) Calculate the Fermi energy.(b) If the experimental value of E_F is 12 eV , what is the electron effective mass in aluminum? [Aluminum is trivalent].

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Analyze the circuit shown in the figure below. Determine the logic function performed by the circuit by making and justifying the appropriate truth table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.